

Comparison of Machine Learning Models for Predicting Depression in Adults Using Behavioral Biomarkers

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Abstract—This document is a model and instructions for L^AT_EX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.

Index Terms—machine learning, depression, predictive modeling, digital biomarkers, BRFS

I. INTRODUCTION

Major depressive disorder represents one of the most critical public health challenges globally, profoundly affecting patients' quality of life and straining healthcare systems. Traditionally, clinical diagnosis and treatment selection have relied heavily on subjective psychiatric evaluations and a reactive, "trial and error" paradigm [1], [2]. As the global burden of mental health disorders grows, there is an urgent need to transition toward proactive, preventative, and personalized psychiatric care.

In recent years, Artificial Intelligence (AI) has emerged as a transformative tool in psychiatric research and decision-making [3]. Cutting-edge studies have successfully applied machine learning (ML) and deep learning algorithms to predict depression using complex clinical biomarkers. For instance, predictive models have been trained on high-dimensional gray-matter structures from brain magnetic resonance imaging (MRI) [4], blood DNA methylation profiles [1], and multimodal sensing networks that integrate audio, video, and text data [5]. Furthermore, AI-powered Clinical Decision Support Systems (CDSS) have been developed to predict psychotherapy remission rates [6], demonstrating clinical feasibility and acceptability without negatively impacting the physician-patient relationship during real-world consultations [2], [7].

Despite these significant technological advances, a notable gap remains in the translational application of these predictive

models to routine clinical practice [3]. Many state-of-the-art diagnostic algorithms depend on expensive, highly specialized, or invasive data sources (e.g., neuroimaging or genomics) that are not readily available in standard primary care or community settings. However, few studies have deeply explored the predictive power of highly accessible, self-reported community health data to proactively estimate depression risk. There remains an unresolved need to evaluate how everyday socio-demographic, economic, and behavioral variables can act as reliable "digital biomarkers" on a large population scale.

To address this knowledge gap, this study aims to estimate the risk of depression in adults through a comparative evaluation of robust machine learning classifiers, utilizing everyday health variables and associated risk factors. By transforming static epidemiological data into a dynamic predictive tool, this research seeks to identify which behavioral and psychosocial biomarkers exhibit the strongest predictive influence on depression. Ultimately, the contribution of this work lies in demonstrating the viability of using accessible, non-invasive community data to support early risk detection and personalized psychiatric interventions.

The remainder of this article is organized as follows: Section II details the methodology, including the dataset extraction, feature ranking, and the configuration of the machine learning algorithms. Section III presents the experimental results and performance metrics. Section IV provides a discussion of the clinical implications of the findings. Finally, Section V concludes the paper and outlines future research directions.

II. METHOD

A. Study Approach and Design

This article develops a predictive and comparative study based on a quantitative and descriptive approach. The primary goal of the research is to estimate the risk of depression in adults through a comparative evaluation of machine learning

algorithms, identifying which socio-demographic and behavioral biomarkers exert the greatest predictive influence. A non-experimental, cross-sectional design is proposed, analyzing retrospective epidemiological databases to construct a clinical classification model [1]

B. Development Methodology

For the execution of the study, the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology [8] was adopted. This framework was chosen due to its cyclical and industry-standard structure, which is ideal for the systematic development of predictive models in the healthcare domain. The analytical tools were based on the Python programming language, utilizing specialized libraries such as Pandas [9] for tabular manipulation via `pyreadstat` and Scikit-learn [10] for algorithmic modeling.

C. Data Source and Extraction Methodology

The analyzed data originate from the *Behavioral Risk Factor Surveillance System* (BRFSS, 2024 cycle) a large-scale epidemiological surveillance survey conducted by the Centers for Disease Control and Prevention (CDC) [11]. The original dataset contains 455,006 records and 301 individual variables representing a wide array of population health metrics.

To ensure computational efficiency and focus on the most relevant information, data extraction was managed using the `pyreadstat` library, selectively loading only the required columns directly from the `.XPT` file. This extraction methodology prevents memory overload and speeds up the preprocessing pipeline. For the feature-selection step, a 100,000-row subsample was used to keep the RF and mutual-information rankings tractable; the selected set was then evaluated on the full 455,006-row dataset.

D. Study Variables and Target Definition

The definitive clinical target variable for this predictive model is `ADDEPEV3`, which represents whether a patient has ever been told by a health professional that they have a depressive disorder [11]. During the data preparation pipeline, this categorical variable is binarized, mapping positive diagnoses to 1 and negative diagnoses to 0, while non-responses are discarded to maintain the integrity of the supervised learning process. While other depression-related variables exist in the dataset (e.g., `_MENT14D`), `ADDEPEV3` was selected as the primary target due to its direct and definitive clinical diagnostic value.

Feature selection was performed by training a Random Forest classifier on a 100,000-row subsample over the full 301-column universe, ranking variables by Gini importance. Eleven variables were excluded from the universe (Table I); the canonical predictor set is the top-30 by RF importance, with two documented substitutions: `_MENT14D` (RF rank 2) was excluded because it is a near-duplicate of the target `ADDEPEV3`, and `_STSTR` (RF rank 30) was excluded because it is a sampling stratum rather than a predictor. `_PACKYRS` (RF rank 36) and `MEDCOST1` (RF rank 47) were substituted

in to replace them. The literal next two RF ranks (31 and 32, importances 0.00605 and 0.00572) were not used: rank 31 is the mean-encoded form of `_RACE`, which is represented separately, and rank 32 (`_CASTHM1`) is a redundant legacy variant of `_ASTHMS1` at rank 17. The 100,000-row subsample was used for the feature-selection step alone (Random Forest fitting, mutual information, and permutation importance are not tractable on 455,006 rows for the full universe); the chosen predictor set was then trained and evaluated on the full 455,006-row dataset in the 5-fold cross-validation. The chosen set contains 30 raw predictors (8 numeric, 4 ordinal, 18 categorical), with `_RACE` mean-encoded separately; the model therefore receives 31 input features per row. The relevant RF ranking is included in Appendix V.

TABLE I
VARIABLES EXCLUDED FROM THE FEATURE-SELECTION UNIVERSE

Variable	Category	Reason for exclusion
ADDEPEV3	target	Lifetime depression diagnosis; the prediction target.
_STATE	geography	U.S. state of residence.
SEQNO	identifier	Respondent sequence number.
_PSU	identifier	Primary sampling unit.
_STRWT	design weight	Design stratum weight.
_WT2RAKE	design weight	Final raking weight.
_LLCPWT	design weight	Landline-cellphone design weight.
_LLCPWT2	design weight	Design weight (LLCP 2024).
_CLLCPWT	design weight	Combined landline/cellphone weight.
IDATE	structural	Interview date.
WEIGHT2	redundant	Reported weight in pounds; redundant with <code>WTKG3</code> .

As detailed in Table II, the chosen set spans cognitive and psychosocial states (`DECIDE`, `DIFFALON`, `SDLONELY`, `LSATISFY`), self-reported mental and physical health days (`MENTHLTH`, `POORHLTH`, `PHYSHLTH`, `_PHYS14D`, `_RFHLTH`, `GENHLTH`), demographic and anthropometric variables (`_AGE80`, `_AGEG5YR`, `_SEX`, `SEXVAR`, `CELLSEX3`, `_BMI5`, `HEIGHT3`, `WTKG3`), socioeconomic and behavioural indicators (`EMPLOY1`, `MEDCOST1`, `ECIGNOW3`, `_PACKYRS`), and chronic-condition and screening history (`ASTHMA3`, `_LTASTH1`, `_ASTHMS1`, `HAVARTH4`, `_DRDXAR2`, `HIVTST7`, `_AIDTST4`, `HIVTSTD3`).

E. Data Preparation and Processing

Data cleaning constituted a critical phase to avoid mathematical biases. According to the BRFSS codebook [11] specific numerical codes like 7, 8, 9, 14, 77, 99, and 9999 represent non-responses (e.g., “Don’t know” or “Refused”). These were mapped to `NaN` to prevent models from interpreting them as continuous weights. Additionally, values of 88 in `PHYSHLTH`, `MENTHLTH`, and `POORHLTH` (meaning “None”) were explicitly mapped to 0 to preserve their clinical meaning.

Categorical variables were processed using smoothed mean encoding and One-Hot Encoding, while numerical variables were normalized using a Standard Scaler. The single `_RACE` column is mean-encoded rather than one-hot-encoded: a single regularised numeric effect is easier to interpret in isolation and avoids one-hot encoding amplifying racial-disparity.

TABLE II
TOP-30 PREDICTORS OF THE CHOSEN SET (RF-RANKED)

Variable	Type	Brief Description
MENTHLTH	numeric	Number of days with poor mental health
POORHLTH	numeric	Days where poor health limited usual activities
DECIDE	categorical	Difficulty concentrating, remembering, or making decisions
_PHYS14D	categorical	14+ days of poor physical health
PHYSHLTH	numeric	Number of days with poor physical health
GENHLTH	ordinal	Self-reported general health
SDLONELY	ordinal	Frequency of feeling lonely / socially isolated
DIFFALON	categorical	Difficulty doing errands alone
_RFHLTH	categorical	Self-reported good or better general health
_SEX	categorical	Biological sex
SEXVAR	categorical	Sex variable
ECIGNOW3	categorical	Current e-cigarette use
_AGE80	numeric	Imputed age (top-coded at 80)
EMPLOY1	categorical	Current employment status
ASTHMA3	categorical	Ever diagnosed with asthma
_ASTHMS1	categorical	Computed current asthma status
_AGE5YR	ordinal	Age, 5-year bands
_BMI5	numeric	Computed Body Mass Index
LSATISFY	ordinal	Self-reported life satisfaction
HIVTST7	categorical	Ever tested for HIV
_LTASTH1	categorical	Computed lifetime asthma status
_DRDXAR2	categorical	Computed diagnosed arthritis status
CELLSEX3	categorical	Sex variable from the cellular-phone sample
_AIDTST4	categorical	Computed indicator: ever been tested for HIV
HIVTSTD3	categorical	Date / recency of last HIV test
HEIGHT3	numeric	Reported height
HAVARTH4	categorical	Ever diagnosed with arthritis
WTKG3	numeric	Computed weight in kg
_PACKYRS	numeric	Computed pack-years of smoking
MEDCOST1	categorical	Could not see a doctor because of cost
Plus _RACE (mean-encoded; encoded column at RF rank 31, importance 0.00605; see Section II.E).		
<i>Note.</i> Variables are listed in RF-importance order on the 100k subsample. _MENT14D (RF rank 2) and _STSTR (RF rank 30) are excluded by documented substitution; _PACKYRS and MEDCOST1 substitute in. The full RF ranking is provided in Appendix V.		

patterns in downstream predictions, while the smoothing prior ($\alpha = 10$) prevents overfitting on small groups. Of the seven race-ethnicity variables available in BRFSS 2024 (_RACE, _RACEGR3, _RACEG21, _RACEPRV, _IMPRACE, _MRACE1, _CRACE1), _RACE was selected as the canonical CDC reporting standard; the other variables are alternative cardinalities or non-Hispanic-specific cuts of the same construct and would not change the model materially. Race was included as a feature for two reasons. Empirically, the encoded column has RF rank 31 with Gini importance 0.00605, placing it above all variables below the top-30 cutoff; the model is marginally better with it than without. Clinically, depression prevalence is known to vary systematically by race in US epidemiological data due to a combination of social determinants (discrimination, healthcare access) and measurement effects (underdiagnosis in some groups, overdiagnosis in others); the encoded form lets the model learn this baseline. We note that the inclusion of race in a depression-prediction model has

known fairness implications and that the fairness properties of the model are not separately validated in this study; the trade-off is left to the deploying institution's judgement. Finally, missing values were imputed using the median for numeric/ordinal data and the mode for categorical data.

F. Dataset Partitioning and Balancing Justification

Given the highly imbalanced nature of epidemiological datasets (where healthy individuals vastly outnumber those with depression), a 5-fold Stratified Cross-Validation (StratifiedKFold) technique was implemented over the full 455,006-row dataset. This methodology ensures that the exact proportion of the target class is preserved across all training and validation folds, allowing for robust evaluation.

Class imbalance was addressed with `class_weight="balanced"` across all five models. No resampling technique was applied. Synthetic resampling (SMOTENC) was also tried on a subsample and was discarded for the same reason: it did not improve any of the worst-case metrics and increased training time without changing the model ranking.

G. Data Modeling and Hyperparameter Justification

Five machine learning architectures were deployed. The five models were configured with the hyperparameters shown in Table IV; all five used `class_weight="balanced"` and `random_state=42` for reproducibility. An additional hyperparameter-tuning loop was attempted on a subsample and was discarded because it produced no meaningful gain on F1 or PR-AUC and did not change the model ranking.

- **Random Forest:** an ensemble of decision trees trained on bootstrapped samples with random feature subsets, aggregated by majority vote.
- **Linear SVC:** a linear support-vector classifier wrapped in `CalibratedClassifierCV` to expose probability estimates from the SVM margins.
- **Logistic Regression:** a linear model using a logistic link, with the `newton-cholesky` solver for efficient fitting on large dense data.
- **XGBoost:** a depth-wise gradient-boosting ensemble with regularized objective.
- **LightGBM:** a leaf-wise gradient-boosting ensemble optimized for high-dimensional, mixed-type data.

H. Evaluation Metrics

Instead of relying solely on baseline accuracy, the models were evaluated using a comprehensive suite of metrics appropriate for imbalanced clinical data, avoiding theoretical mathematical derivations in favor of applied clinical relevance:

- **F1-Score & Recall:** Recall (Sensitivity) measures the model's ability to identify true depression cases, minimizing dangerous false negatives. The F1-Score provides the harmonic mean between precision and recall, ensuring a balanced performance.
- **ROC-AUC & PR-AUC:** The Area Under the Receiver Operating Characteristic (ROC-AUC) curve evaluates the

model’s discriminative threshold. Given the class imbalance, the Precision-Recall AUC (PR-AUC) was also calculated to provide a stricter performance penalty for false positives.

- **Brier Score & Confusion Matrix:** The Brier score measures the exactness of the probabilistic outputs, while the accumulated Confusion Matrix provides a direct visualization of true positive, true negative, false positive, and false negative predictions across the 5 folds.

In addition to the standard probability metrics, the per-model decision threshold was tuned out-of-fold by maximizing F1; alternatives such as Youden’s J statistic and a cost-sensitive criterion (with a user-defined cost ratio) are also applicable for screening and cost-sensitive applications. The reported `*@thr` columns show metrics at the chosen threshold, and the `threshold` column reports the threshold itself. In this study we report the F1-tuned threshold (5-fold mean).

I. Implementation Environment

The system was implemented using a high-performance computational environment executed via Python scripts. Mathematical processing relied on NumPy [12] and Pandas [9] modules, while the machine learning experimentation framework depended on the Scikit-Learn [10] ecosystem.

III. RESULTS

The findings of the predictive modeling experiment are presented based on the evaluation of five machine learning architectures over a 5-fold stratified cross-validation process on the full 455,006-row dataset. The algorithms were evaluated on their ability to detect the ADDEPEV3 target using the 30 socio-behavioral biomarkers of the chosen predictor set. Thresholds are tuned per model out-of-fold to maximize F1; the `*@thr` columns report metrics at the chosen threshold, while the unsubscripted accuracy is reported at the 0.5 default threshold.

Table III summarizes the performance metrics for each model. XGBoost achieves the highest unsubscripted accuracy (0.840) and the lowest Brier score (0.116), indicating the best-calibrated probability estimates. XGBoost and LightGBM are tied for the highest ROC-AUC (0.843) and PR-AUC (0.635). Under the F1-tuned threshold, XGBoost and LightGBM also tie for the highest F1 (0.597): LightGBM is marginally ahead in recall (0.652 vs 0.644) and XGBoost is marginally ahead in precision (0.556 vs 0.552).

LinearSVC shows the largest swing from the 0.5-threshold to the F1-tuned threshold: at 0.5 it has the second-highest accuracy on the validation set (0.833, behind only XGBoost at 0.840) but very low recall (0.378). F1-tuning moves the threshold down to 0.248 and raises recall to 0.638 at the cost of precision, giving F1 0.577. The linear baseline and the tree-based bagging model (Random Forest) reach similar F1 at the tuned threshold (0.578 and 0.581 respectively) and lag the gradient-boosting pair on ROC-AUC, PR-AUC, and Brier. Appendix Table ?? lists all five threshold-dependent metrics at the default 0.5 threshold for direct comparison.

TABLE III
MODEL EVALUATION RESULTS (5-FOLD MEAN, FULL 455,006-ROW DATASET)

Model	Acc@0.5	Acc@thr	Prec@thr	Rec@thr	F1@thr	Spec@thr	ROC-AUC	PR-AUC	Brier	Threshold
Random Forest	0.769	0.803	0.528	0.646	0.581	0.845	0.831	0.612	0.171	0.553
Linear SVC	0.833	0.802	0.526	0.638	0.577	0.846	0.825	0.607	0.121	0.248
Logistic Regression	0.783	0.801	0.523	0.646	0.578	0.842	0.826	0.608	0.164	0.545
XGBoost	0.840	0.816	0.556	0.644	0.597	0.862	0.843	0.635	0.116	0.304
LightGBM	0.775	0.814	0.552	0.652	0.597	0.858	0.843	0.635	0.157	0.614

TABLE IV
MODEL HYPERPARAMETERS

Model	Hyperparameters
Random Forest	n_estimators=300, max_depth=12, n_jobs=-1, random_state=42, class_weight="balanced"
Linear SVC	LinearSVC(dual="auto", max_iter=2000, class_weight="balanced") wrapped in CalibratedClassifierCV(cv=3, method="sigmoid")
Logistic Regression	solver="newton-cholesky", max_iter=2000, random_state=42, class_weight="balanced"
XGBoost	n_estimators=300, max_depth=6, learning_rate=0.1, n_jobs=-1, class_weight="balanced"
LightGBM	n_estimators=300, learning_rate=0.1, n_jobs=-1, random_state=42, class_weight="balanced"

The accumulated confusion matrices detail the specific classification behavior across the evaluated folds. As observed in Figure 1, LightGBM correctly identifies the largest share of true-positive depression cases at the F1-tuned threshold while keeping the false-negative count low. In contrast, while XGBoost (Figure 2) reaches a comparable true-positive count, its higher threshold-driven precision is paired with a higher true-negative count, reflecting its advantage in calibration.

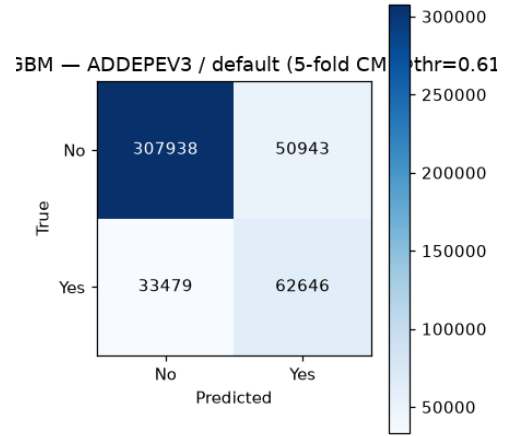


Fig. 1. Confusion Matrix for LightGBM (5-fold Cross-Validation, F1-Tuned Threshold)

IV. DISCUSSION

The primary objective of this study was to estimate the risk of depression in adults by identifying the most influential socio-behavioral biomarkers and evaluating various predictive

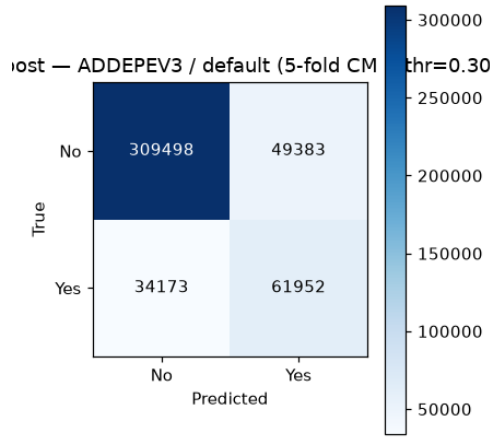


Fig. 2. Confusion Matrix for XGBoost (5-fold Cross-Validation, F1-Tuned Threshold)

algorithms. The results indicate that gradient boosting frameworks—specifically LightGBM and XGBoost—are highly effective at modeling complex, non-linear epidemiological data, surpassing traditional linear approaches on both ranking and calibration metrics.

A critical finding of this research is the importance of threshold choice when reporting results on imbalanced clinical data. Under a fixed 0.5 threshold, the SVM and gradient-boosting models favour the majority class and miss a large share of positive cases (LinearSVC recall 0.378, XGBoost recall 0.441). F1-tuning the threshold per model closes this gap, with all five models reaching $F1 \in [0.577, 0.597]$ and $\text{recall} \in [0.638, 0.652]$. The remaining separation is in calibration: XGBoost and LightGBM share the top ROC-AUC (0.843) and PR-AUC (0.635), and XGBoost has the best Brier score (0.116), making it the strongest single candidate when well-calibrated probabilities are required. LightGBM reaches the same ROC-AUC with marginally higher recall (0.652 vs 0.644) at the F1-tuned threshold, consistent with LightGBM’s leaf-wise design for high-dimensional, mixed-type data. LinearSVC and Logistic Regression are competitive on F1 at the tuned threshold but trail on Brier and on PR-AUC.

Despite these promising results, the study presents methodological limitations. The reliance on the BRFSS dataset implies that the input variables are self-reported, making them susceptible to recall bias and subjective underreporting of symptoms. Furthermore, even after F1-tuning, the precision across all models remains moderate (0.52–0.56), reflecting an inherent challenge when working with highly imbalanced epidemiological data where symptoms overlap with other conditions.

Future research should focus on mitigating this residual false-positive rate through more advanced resampling or cost-sensitive learning. In addition, future work should evaluate the same predictors on a target measuring *recent* depressive symptoms, in a setup where `_MENT14D` is excluded from

predictors to avoid target–predictor overlap. This would test transfer from lifetime-diagnosis to acute-risk prediction.

V. CONCLUSIONS

This study successfully demonstrated that machine learning algorithms can reliably estimate the risk of lifetime depression in adults utilizing accessible, non-invasive behavioral and psychosocial data. By identifying key digital biomarkers, this research confirms that everyday variables—such as socioeconomic status, mental and physical health self-reports, cognitive function, and behavioral indicators—hold substantial predictive power, reducing the reliance on costly clinical tests.

Among the evaluated architectures, XGBoost and LightGBM are tied at the top of the metric table. XGBoost is the strongest single model when well-calibrated probabilities are required (Brier 0.116, ROC-AUC 0.843, $F1@thr$ 0.597), while LightGBM reaches the same ROC-AUC with marginally higher recall (0.652 vs 0.644) at the F1-tuned threshold.

The developed predictive framework offers direct practical applications for healthcare systems. These models can be seamlessly integrated into Clinical Decision Support Systems (CDSS) within primary care settings to automatically flag at-risk individuals using standard patient intake forms. Future investigations should expand the feature set and focus on longitudinal validations to further refine the precision of the algorithm in diverse clinical environments.

APPENDIX

Table V lists the Random Forest Gini-importance top-30 of the chosen predictor set, plus the two substituted-in variables from later RF ranks. Rows marked with a single asterisk are excluded from the chosen set; rows marked with a double asterisk are substituted in.

TABLE VI
MODEL PERFORMANCE AT THE DEFAULT 0.5 THRESHOLD (5-FOLD
MEAN, FULL 455,006-ROW DATASET)

Model	Acc@0.5	Prec@0.5	Rec@0.5	F1@0.5	Spec@0.5
Random Forest	0.769	0.469	0.728	0.571	0.780
Linear SVC	0.833	0.693	0.378	0.489	0.955
Logistic Regression	0.783	0.490	0.691	0.573	0.808
XGBoost	0.840	0.691	0.441	0.538	0.947
LightGBM	0.775	0.480	0.750	0.585	0.782

Note. All five metrics are evaluated at the default 0.5 decision threshold (no per-model tuning). Compare against Table III which reports the same metrics at the F1-tuned per-model threshold; the contrast illustrates the F1-vs-accuracy trade-off for imbalanced classification. Threshold-independent metrics (ROC-AUC, PR-AUC, Brier) are unchanged and are not repeated here.

TABLE V
RANDOM FOREST TOP-30 BY GINI IMPORTANCE (CANONICAL)^a

Rank	Variable	Importance
1	MENTHLTH	0.15857
2	_MENT14D*	0.14782
3	POORHLTH	0.07117
4	DECIDE	0.05926
5	_PHYS14D	0.02364
6	PHYSHLTH	0.02300
7	GENHLTH	0.01447
8	SDLONELY	0.01446
9	DIFFALON	0.01355
10	_RFHLTH	0.01305
11	_SEX	0.01217
12	SEXVAR	0.01164
13	ECIGNOW3	0.01133
14	_AGE80	0.01007
15	EMPLOY1	0.00950
16	ASTHMA3	0.00900
17	_ASTHMS1	0.00887
18	_AGEG5YR	0.00877
19	_BMI5	0.00876
20	LSATISFY	0.00815
21	HIVTST7	0.00791
22	_LTASTH1	0.00717
23	_DRDXAR2	0.00714
24	CELLSEX3	0.00712
25	_AIDTST4	0.00676
26	HIVTSTD3	0.00667
27	HEIGHT3	0.00649
28	HAVARTH4	0.00638
29	WTKG3	0.00637
30	_STSTR*	0.00632
36	_PACKYRS**	0.00502
47	MEDCOST1**	0.00380

^a*Note.* *Excluded from the chosen set by documented substitution: _MENT14D (rank 2) as a near-duplicate of the target, _STSTR (rank 30) as a sampling stratum. **Substituted in from later RF ranks to replace the excluded rows: _PACKYRS (rank 36), MEDCOST1 (rank 47).

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